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Research paper

Interoceptive sensibility mediates emotional dysregulation: Insights from individuals with Bipolar II Depression

Gaspare Alfi [a,*], Giovanna Grenno [a], Valerio Caruso [a], Andrea Piarulli [a], Laura Palagini [b], Giulio Perugi [c], Angelo Gemignani [a,b]

[a] Department of Surgical, Medical and Molecular Pathology and Critical Care Medicine, University of Pisa, Pisa, Italy
[b] Department of Neuroscience, University of Pisa Hospital, Pisa, Italy
[c] Department of Clinical and Experimental Medicine, University of Pisa, Pisa, Italy

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ABSTRACT

Emotion dysregulation (ED), a core feature of Bipolar Disorder (BD), contributes to symptom severity, mood instability, and reduced quality of life. Emerging evidence suggests that interoception—the ability to perceive and integrate internal bodily signals—may play a pivotal role in shaping emotional experience. However, the relationship between interoceptive sensibility and ED in BD remains poorly understood.

This study investigated whether interoceptive sensibility mediates ED in individuals with Bipolar II Disorder during a depressive episode (BD2-D). Sixty inpatients with BD2-D and 36 healthy controls completed a psychometric assessment including the Difficulties in Emotion Regulation Scale (DERS), the Multidimensional Assessment of Interoceptive Awareness (MAIA), the Beck Depression Inventory-II (BDI-II), and the Self-Rating Anxiety Scale (SAS).

Compared to controls, patients exhibited significantly higher ED across all DERS components (p-value <0.01), and lower interoceptive sensibility, both in total scores and in six of eight MAIA components ($p < 0.005$). Partial correlations showed strong negative associations between interoceptive sensibility and ED, particularly in self-regulation and trust in bodily signals. Mediation analysis indicated that interoceptive sensibility significantly mediated the relationship between BD2-D and ED (indirect effect $\beta = 16.18$, 95 % CI [8.54, 27.54]).

These findings highlight the pivotal role of interoceptive sensibility in emotional regulation in BD2-D. Impaired integration of bodily signals may jeopardize self-regulation and increase emotional vulnerability. Interoception may therefore represent a promising target for intervention in mood disorders.

1. Introduction

Bipolar Disorder (BD) is a severe and chronic psychiatric condition characterized by recurrent episodes of mania, hypomania, depression, and mixed states, affecting approximately 1 % of the global population (Rowland and Marwaha, 2018). These mood fluctuations significantly impair daily functioning and quality of life, contributing to high rates of disability, mortality, and suicide risk (Nierenberg et al., 2023; Oliva et al., 2023). Among the core features of BD, Emotion Dysregulation (ED) has emerged as a fundamental mechanism influencing both the onset and the course of the disorder (De Prisco et al., 2023).

ED encompasses a range of definitions, terms, and manifestations, reflecting the heterogeneity with which the construct is understood and applied. At its core, however, it is commonly described as involving rapid oscillations of intense emotional states and a diminished capacity to regulate these fluctuations or their behavioral outcomes (Brancati et al., 2023). Hence, ED represents a multidimensional and trans-diagnostic construct, referring to a patterns of emotional experience and expression that interfere with the ability to engage in goal-directed behavior (Thompson, 2019). In particular, ED includes difficulties in controlling impulsive behaviors or modulating emotional responses to negative emotions (Gratz and Roemer, 2004), abrupt or excessively slow changes in emotions, and a slower return to an emotional baseline (Carpenter and Trull, 2013). In individuals with BD, ED is also associated with atypical attentional allocation to emotional stimuli (García-Blanco et al., 2015; Henry, 2012).

From a neurobiological perspective, ED in BD arises from multi-level abnormalities in both fronto-limbic circuits and large-scale brain

* Corresponding author.
E-mail address: gaspare.alfi@med.unipi.it (G. Alfi).

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networks, which play a crucial role in emotional regulation (Bi et al., 2022). At the core level, disruptions in the fronto-limbic circuit—where a hyperactive amygdala is insufficiently regulated by the ventromedial prefrontal cortex (vmPFC) and the anterior cingulate cortex (ACC)—lead to heightened emotional reactivity and poor affective modulation. (Lawrence et al., 2004; Radaelli et al., 2015; Strakowski et al., 2012; Townsend et al., 2013; Wessa and Linke, 2009). At higher levels, dysregulations in large-scale brain networks further impair emotional control. Specifically, hyperactivity in the Default Mode Network (DMN) and Salience Network (SN), combined with reduced activity in the Central Executive Network (CEN), results in severe deficits in cognitive control over emotional responses, contributing to mood liability (Ellard et al., 2018; Goya-Maldonado et al., 2016; Zarp Petersen et al., 2022; Zovetti et al., 2020).

From a clinical standpoint, ED exacerbates manic and depressive symptom severity, and overall illness burden (De Prisco et al., 2023). During manic episodes, ED contributes to heightened emotional reactivity, impulsivity, and increased sensitivity to reward, likely driven by amygdala hyperactivity and reduced prefrontal inhibitory control (Mesbah et al., 2023; Phillips and Swartz, 2014). On the other hand, during depressive episodes, ED is characterized by blunted affect, rumination, and impaired emotional modulation, associated with hypoactive limbic responses and DMN hyperactivity (Rey et al., 2021). In mixed states, ED results in heightened emotional distress and rapid mood fluctuations, reflecting simultaneous hyperactivity in emotional processing-related brain networks, weakness in cognitive control (M'Bailara et al., 2012), and dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis (Fitzgerald et al., 2018). Even in euthymia, persistent fronto-limbic dysconnectivity suggests a trait-like vulnerability, predisposing individuals to future mood episodes (Bi et al., 2022). Understanding the underlying mechanisms of ED in BD is therefore crucial for predicting clinical outcomes and developing tailored interventions.

In mental health, an emerging field of investigation is represented by interoception, which refers to the organism's ability to perceive, interpret and regulate internal signals of the body, both from a conscious (e.g. hunger, thirst, heartbeat, muscle tension, visceral pain) and an unconscious level (e.g. variations in neurotransmitters, hormones, receptors) (Chen et al., 2021). This process occurs through the interplay between brain networks anchored to specific areas (insula, anterior cingulate cortex, vagus nerve), physiological signals of the body (e.g. receptor stretching, cellular turnover), and experience in the world (Chen et al., 2021; Khalsa et al., 2018).

According to the Embodied-Predictive Interoception Coding (EPIC) model, through these mechanisms, the organism is able to predict internal and external requests in a Bayesian way and generate appropriate behavioral responses (Barrett and Simmons, 2015). Alterations at this level are implicated in the pathophysiology of a wide range of psychiatric disorders, including anxiety, mood disorders, schizophrenia, and addiction (Bonaz et al., 2021). In this sense, emotions do not arise in isolation from the body, nor are they reducible to cognitive interpretation alone. Rather, they emerge from the integration of distributed interoceptive signals – cardiac, respiratory, and visceral – into unified experiential states (Critchley and Garfinkel, 2017). When interoception is dysregulated – whether through amplification, attenuation, or loss of signals integration – the resulting perturbations can manifest as affective instability, heightened stress reactivity, and impaired adaptive behavior (Khalsa et al., 2018).

Building on these insights, ED and interoceptive impairments could be conceptualized as elements of a broader transdiagnostic domain spanning several psychiatric conditions.

Bipolar Disorder offers a persistent, internally driven form of affective instability, likely reflecting a trait-like neurobiological vulnerability (De Prisco et al., 2023; Durdurak et al., 2025). This trait orientation contrasts with the more context-dependent forms of emotion dysregulation observed in other conditions (Carpenter and Trull, 2013; Shaw et al., 2014).

For instance, Borderline Personality Disorder (BPD), Attention-Deficit/Hyperactivity Disorder (ADHD), and atypical depression have all been associated with both emotion dysregulation and interoceptive dysfunction (Carpenter and Trull, 2013; Flasbeck et al., 2020; Hu et al., 2023; M. Bruton et al., 2025; Shaw et al., 2014), though likely with differing profiles and mechanisms.

In BPD and ADHD, emotion dysregulation tends to manifest as a rapid, reactive response to external stimuli and is closely linked to interpersonal instability or attentional dyscontrol (Martel, 2009; Sebastian et al., 2013; Shaw et al., 2014). In atypical depression, ED seems related to rejection sensitivity and somatic amplification (Silverstein et al., 2006).

Thus, this study adopts a real-world approach by focusing on Bipolar II Disorder inpatients hospitalized for depressive episodes, the most common reason for admission in our clinical setting, drawing on both the high availability of this patient group and the specific opportunities that BD offers for investigating the potential mediating role of interoceptive sensibility in emotion dysregulation.

2. Methods

2.1. Participants

The current study involved a subsample of participants ($n = 94$) from an ongoing research project involving patients with Bipolar Disorder type I or II. All recruited patients were admitted to the Psychiatry Unit of the University of Pisa, Italy, for a depressive episode, from July 2024 to February 2025.

A group of healthy participants balanced for sex and age was recruited from university personnel. Eligible individuals were aged between 18 and 65 years and agreed to provide informed consent for the study. All participants underwent evaluation using a set of questionnaires, which included the Structured Clinical Interview for DSM-5-TR (SCID-5) (First et al., 2017) to assess the presence of current or lifetime psychiatric diagnoses.

The exclusion criteria for healthy controls were as follows: a) a past or current diagnosis of mental disorders; b) a history of chronic medical conditions; c) regular use of alcohol or drugs; d) age < 18 e) failure to complete the questionnaires; f) pregnancy; g) any other condition deemed by the investigators to warrant exclusion.

The study adhered to the Declaration of Helsinki, and both inpatients and healthy subjects provided written informed consent before being enrolled. The study was approved by the local ethical committee titled "CRONOBIO" n° 26,869.

2.2. Psychiatric evaluation

Psychiatric diagnoses were evaluated according to DSM-5-TR criteria (American Psychiatric Association, 2022) using the Structured Clinical Interview for Axis I Disorders (SCID-5) (First et al., 2017). Interviews were conducted by trained psychiatrists, applying clinical judgment throughout the process.

2.3. Psychometric assessment

The psychometric assessment, conducted by a trained psychologist, was performed upon admission, while patients were still on their initial pharmacological regimen and before the introduction of the new treatment. The assessment comprised the following questionnaires:

Difficulties in Emotion Regulation Scale (DERS) (Gratz and Roemer, 2004; Sighinolfi et al., 2010). The DERS is a 36-item self-report instrument designed to assess emotion dysregulation, using a Likert-type scale ranging from 1 (almost never) to 5 (almost always). The total score, obtained by summing all items, ranges from 36 to 180, with higher scores indicating greater overall difficulties in emotion

regulation. The questionnaire also comprises six subscales, each reflecting distinct facets of emotion regulation difficulties: (1) Nonacceptance of Emotional Responses, assessing negative secondary reactions to distress; (2) Difficulties in Goal-Directed Behavior, measuring challenges in maintaining focus and task completion during negative emotional states; (3) Impulse Control Difficulties, evaluating problems in behavioral self-control when emotionally distressed; (4) Lack of Emotional Awareness, assessing difficulties in identifying or attending to emotional states; (5) Limited Access to Regulation Strategies, reflecting the belief that few effective strategies are available when upset; and (6) Lack of Emotional Clarity, measuring confusion about the nature of one's emotions. The DERS has demonstrated good psychometric properties and is considered a transdiagnostic measure of emotion regulation difficulties across various psychiatric conditions.

The Multidimensional Assessment of Interoceptive Awareness (MAIA) (Calì et al., 2015; Mehling et al., 2012). The MAIA is a widely used self-report questionnaire designed to assess multiple dimensions of interoceptive sensibility. It consists of 32 items rated on a Likert scale ranging from 0 (never) to 5 (always), which together yield eight distinct components: Noticing, Not-Distracting, Not-Worrying, Attention Regulation, Emotional Awareness, Self-Regulation, Body Listening, and Trust. Specifically, 1) *Noticing* refers to the awareness of bodily sensations, including discomfort, comfort, and neutral sensations. 2) *Not-Distracting* assesses the tendency not to ignore or distract oneself from sensations of pain or discomfort, whereas 3) *Not-Worrying* evaluates the tendency not to worry about or experience emotional distress in response to such sensations. 4) *Attention Regulation* reflects the ability to sustain and control attention toward bodily sensations. 5) *Emotional Awareness* measures the recognition of the connection between bodily sensations and emotional states. 6) *Self-Regulation* captures the ability to regulate distress by focusing attention on bodily sensations. 7) *Body Listening* describes actively listening to the body for insight. Lastly, 8) *Trusting* assesses the experience of one's body as safe and trustworthy. In the present study, we also computed a composite score by summing all 32 items of the MAIA, referred to here as "MAIA Total Score".

Beck Depression Inventory II (BDI-II) (Beck et al., 1996; Montano and Flebus, 2006). The BDI-II is a 21-question tool designed to evaluate the severity of depressive symptoms in adults and adolescents (Beck et al., 1996). For each question, participants are instructed to select the statement that best reflects how they have felt over the past two weeks. Responses are scored on a 4-point scale ranging from 0 to 3. The overall score ranges from 0 to 63, with higher scores reflecting greater severity of depressive symptoms.

Self-rating Anxiety Scale (SAS) (Conti, 1999; Zung, 1971). The SAS is a 20-item self-assessment tool that measures the severity of anxiety symptoms. Each item is rated on a four-point scale, and the total score indicates the level of anxiety, with higher scores reflecting more severe symptoms.

2.4. Statistical analysis

Data management and inferential statistical analyses were performed using IBM SPSS statistic 25. Hayes' (2018) PROCESS macro for SPSS, version 3.5, was used to assess for mediation (Hayes, 2018).

Group comparisons were conducted based on suitable statistical tests: we used $\chi 2$ test, and Independent Sample *t*-test, where appropriate. The normality of continuous data and the homogeneity of the variance were tested by the Shapiro–Wilk test and Levene's test, respectively.

Qualitative values were reported using numbers and percentages, while mean and standard deviation were used for quantitative values.

For each variable, we derived the effect size (ES) (Cohen, 2013) and its magnitude was interpreted according to Cohen classification: 0.0–0.1 (no effect), 0.2–0.4 (small effect), 0.5–0.7 (intermediate effect), and 0.8–1 (large effect).

Statistical significance was set at $p < 0.05$, and all tests were 2-tailed.

A partial correlation was carried out, and *p*-values <0.05 were adjusted using Holm-Bonferroni correction.

The mediation analysis was conducted using the diagnosis of BD as an endogenous categorical independent variable (1 = presence; 0 = absence), and DERS Total Score as a continuous dependent variable. Instead, MAIA Total Score was introduced as a continuous endogenous mediating variable.

To test for statistical significance and obtain the 95 % bias-corrected confidence interval for the indirect effects, standard maximum likelihood bootstrapping was performed by generating 5000 bootstrap samples for the hypothesized model. If the 95 % confidence interval (CI) does not include zero, this indicates a significant indirect effect at the 0.05 level. All reported regression coefficients are standardized.

3. Results

3.1. Demographics

Out of the initial 94 patients, 14 were excluded due to substance abuse, 10 due to comorbid ADHD, 5 were experiencing psychomotor agitation at the time of assessment, and 5 declined to participate in the study. Consequently, a total of 60 patients diagnosed with Bipolar II Disorder in a depressive phase (BD2-D; 18 males and 42 females, mean age: 41.75 ± 16.59 [mean ± SD]) and 36 healthy individuals (12 males and 24 females, mean age: 38.64 ± 16.87 [mean ± SD]) completed the study. The two groups did not differ in age, sex and educational levels. As expected, BD2-D patients had significantly higher levels of depressive symptoms (BDI-II, *p*-value <0.001) and anxiety (SAS, *p*-value <0.001) than healthy control group (Table 1). At the time of admission, patients with BD2-D were already receiving ongoing pharmacological treatment, which included Lithium (35.00 %), Valproate (36.66 %), Other Antiepileptic Drugs (AEDs; 32.75 %), Selective Serotonin Reuptake Inhibitors (SSRIs; 5.00 %), Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs; 3.33 %), Noradrenergic and Specific Serotonergic Antidepressants (NaSSAs; 1.66 %), Tricyclic Antidepressants (TCAs; 13.33 %), First-Generation Antipsychotics (FGAs; 8.33 %), Second-Generation Antipsychotics (SGAs; 3.33 %), and Benzodiazepines (BDZs; 10.00 %) (Table 1).

Table 1
Demographic and affective characteristics of study participants.

Demographic characteristics	BD2-D (n = 60)	Control group (n = 36)	p-Value
Sex (female)	70 % (n = 42)	66.66 % (n = 24)	0.73
Age	41.75 (±16.59)	38.64 (±16.87)	0.37
Education (years)	13.33 (±2.71)	12.92 (±2.94)	0.49
Mood and anxiety			
BDI II	25.98 (± 5.87)	6.08 (±2.43)	0.001
SAS	41.15 (±7.82)	28.53 (±2.15)	0.001
Medication			
Lithium	35.00 % (n = 21)		
Valproate	36.66 % (n = 22)		
AEDs	32.75 % (n = 19)		
SSRIs	5.00 % (n = 3)		
SNRIs	3.33 % (n = 2)		
NaSSAs	1.66 % (n = 1)		
TCAs	13.33 % (n = 8)		
FGAs	8.33 % (n = 5)		
SGAs	3.33 % (n = 2)		
BDZs	10.00 % (n = 6)		

Abbreviations: BD2-D, Bipolar II Disorder during depressive phase; Beck Depression Inventory-II (BDI II), Self-Rating Anxiety Scale (SAS), Other Antiepileptic Drugs (AEDs), Selective Serotonin Reuptake Inhibitors (SSRIs), Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs), Noradrenergic and Specific Serotonergic Antidepressants (NaSSAs), Tricyclic Antidepressants (TCAs), First-Generation Antipsychotics (FGAs), Second-Generation Antipsychotics (SGAs), Benzodiazepines (BDZs).

3.2. Emotion dysregulation

The difference between the means of the BD2-D and healthy control groups was statistically significant for the total DERS score as well as all six of its components. Specifically, patients with BD2-D showed a significant increase in the DERS Total Score ($p < 0.001$, d = 2.012), as well as in each of its components: Non-Acceptance ($p < 0.001$, d = 1.528), Goals ($p < 0.001$, d = 1.507), Impulse ($p < 0.001$, d = 1.637), Awareness ($p < 0.001$, d = 1.102), Strategies ($p < 0.001$, d = 1.954), and Clarity ($p < 0.001$, d = 2.550) (see Table 2, and Fig. 1a).

3.3. Interoceptive sensibility

The difference between the means of the BD2-D and healthy control groups was statistically significant for the total MAIA score and six out of its eight components. Specifically, patients with BD2-D showed a significant decrease in the MAIA Total Score (*p*-value <0.001, d = 1.168), as well as in the following components: Not-Distracting (*p*-value <0.001, d = 2.313), Not-Worrying (*p*-value <0.001, d = 0.956), Attention Regulation (*p*-value <0.001, d = 0.738), Self-Regulation (p-value =0.004, d = 0.636), Body Listening (*p*-value <0.001, d = 0.718), Trusting (*p*-value <0.001, d = 1.668) (see Table 3, and Fig. 1b).

3.4. Correlation analysis

With the aim to examine the relationship between interoceptive sensibility and ED in BD2-D patients, a total of 64 statistical tests were performed using partial correlation analysis. BDI-II and SAS total scores were included as covariates. After accounting for depressive and anxiety symptoms, the MAIA Total Score, along with seven out of its eight components, showed significant correlations with the DERS total score and five of its six components. Specifically, MAIA TS was negatively correlated with DERS TS ($r = -0.675$, adj *p*-value =0), Not Acceptance ($r = -0.504$, adj p-value =0), Goals ($r = -0.741$, adj p-value =0), Awareness ($r = -0.439$, adj p-value =0.042), Strategies ($r = -831$, adj p-value =0). Noticing was negatively correlated with DERS TS ($r = -0.46$, adj p-value =0), Goals ($r = -0.625$, adj p-value =0), Awareness (r = −0.6, adj p-value =0), Strategies ($r = -0.605$, adj p-value =0). Not-Distracting was positively correlated with Goals ($r = 0.559$, adj p-value =0). Not-Worrying was negatively correlated with Goals ($r = -0.431$, adj p-value =0). Attention Regulation was positively correlated with Goals (r = −0.6, adj p-value =0). Self-Regulation was negatively correlated with DERS TS ($r = -0.592$, adj p-value =0), Non Acceptance ($r = -0.534$, adj p-value =0), Goals ($r = -0.777$, adj p-value =0), Strategies ($r = -0.726$, adj p-value =0). Body Listening was negatively correlated with DERS TS ($r = -0.479$, adj p-value =0), Non Acceptance ($r = -0.562$, adj p-value =0), Awareness ($r = -0.462$, adj p-value =0), Strategies ($r = -0.447$, adj p-value =0). Trusting was negatively correlated with DERS TS ($r = -0.446$, adj p-value =0), Goals ($r = -0.443$, adj p-value =0), Impulse ($r = -0.472$, adj p-value =0), Strategies ($r = -0.646$, adj p-value =0) (Fig. 2).

Table 2
Emotion dysregulation comparison between patients and control group.

DERS components	BD2-D		Control group		p-Value	Effect size
	Mean	(± SD)	Mean	(± SD)		
Not-acceptance	18.60	(± 4.72)	11.94	(± 3.64)	0.001	1.528
Goals	17.28	(± 3.41)	12.25	(± 3.21)	0.001	1.507
Impulse	17.67	(± 5.27)	11.31	(± 2.73)	0.001	1.637
Awareness	16.77	(± 3.49)	13.81	(± 2.05)	0.001	1.102
Strategies	25.87	(± 4.69)	16.64	(± 4.77)	0.001	1.954
Clarity	17.17	(± 3.59)	9.58	(± 2.53)	0.001	2.550
DERS total score	113.23	(± 19.41)	75.33	(± 17.54)	0.001	2.012

Abbreviation: DERS, Difficulties in Emotion Regulation Scale; BD2-D, Bipolar II Disorder during depressive phase; ±SD, ± Standard Deviation; N.A., Not Applicable.

3.5. Mediation analysis

Considering the complete model, BD2-D and the MAIA Total Score together account for 70,30 % of the variance in the DERS Total Score ($R^2 = 0.7030$). Specifically, BD2-D has a direct, positive and significant impact on DERS Total Score (β = 21.51, 95 % CI [LL = 14.30, UL = 28.72]). MAIA Total Score has a direct, negative and significant impact on DERS Total Score (β = 21.51, 95 % CI [LL = 14.30, UL = 28.72]). BD2-D alone accounts for 49.2 % of the variance in DERS Total Score ($R^2 = 0.492$), indicating a substantial effect on increasing ED (β = 37.70, 95 % CI [LL = 29.85, UL = 45.55]). Notably, BD2-D also influences the DERS Total Score indirectly by exerting an effect on the MAIA Total Score (β = 16.18, 95 % CI [LL = 8.54, UL = 27.54]). Indeed, BD2-D account for 30.00 % of the variance in MAIA Total Score ($R^2 = 0.300$), indicating a substantial effect on reducing Interoceptive Sensibility (β = −0.78, 95 % CI [LL = −1.03, UL = −0.53]), which in turn contributes to increased ED (Table 4 and Fig. 3).

4. Discussion

The present study examined the relationship between interoceptive sensibility and ED in patients with Bipolar II Disorder hospitalized for a depressive episode, aiming to explore the potential mediating role of interoception.

As previously noted, ED in bipolar disorder (BD) presents in phase-specific patterns, and our findings add nuance to this framework by highlighting the depressive-phase profile, marked by heightened negative affect and maladaptive strategies such as rumination (Oliva et al., 2025), often linked to amygdala hyperactivity and deficient prefrontal regulation (Wessa and Linke, 2009). Other phases show distinct patterns: hypomania involves poor down-regulation of positive affect, impulsivity, and increased reward sensitivity (Oliva et al., 2025), while mixed states are characterized by rapid shifts between irritability and despair, which markedly increases suicide risk (Tondo et al., 2020). Even in euthymia, subtle deficits in emotional clarity and impulse control persist, indicating a trait-like vulnerability (Van Rheenen et al., 2015). These distinctions do not exclude that the interoceptive mechanisms underlying ED may vary in magnitude and quality across mood states.

Against this background, the current results showed that BD2-D patients, as expected, exhibited significantly higher levels of depressive symptoms (BDI-II) and anxiety (SAS), compared to healthy controls.

Furthermore, our analyses revealed several critical differences in ED and interoceptive sensibility between the two groups.

Firstly, BD2-D exhibited markedly higher DERS total and subscale scores compared to healthy controls, with large effect sizes across all six components (Total DERS: d = 2.01; subscales ranging d = 1.10–2.55). It is noteworthy that the mean DERS total score in patients was 113.26, consistent with the literature, which typically associates scores ≥95 with clinically significant emotional dysregulation and related psychopathology (Erez and Gordon, 2025).

This emerging pattern of emotion dysregulation reflects a lack of ability to monitor, appraise, or control emotional reactions to achieve an appropriate goal-directed behavior, confirming the existing literature (De Prisco et al., 2023; Oliva et al., 2023, 2025).

From a clinical standpoint, although our study focused on BD2-D, prior research has shown that specific DERS dimensions map onto distinct BD psychopathological features. As a matter of fact, impulse control difficulties have been shown to predict manic propensity, while limited access to emotion regulation strategies predicts depressive vulnerability (Van Rheenen et al., 2015). Furthermore, elevated DERS total and subscale scores, particularly in Strategies, Clarity, and Impulse, have been strongly associated with suicidal ideation and past suicide attempts among bipolar patients (Kulacaoglu et al., 2022).

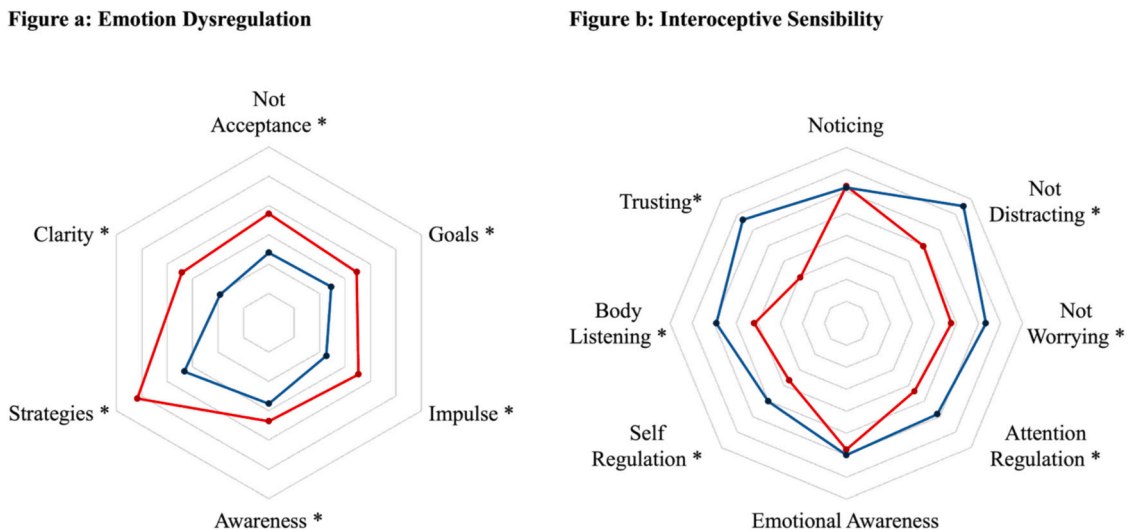

Fig. 1. Psychometric profiles of Emotion Dysregulation and Interoceptive Sensibility in patients and controls.
Note: Figure a, Emotion Dysregulation profile in patients (red line) and controls (blue line); Figure b, Interoceptive Sensibility profile in patients (red line) and controls (blue line); *, Statistical significance (p-value < 0.05). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Interoceptive Sensibility comparison between patients and control group.

MAIA components	BD2-D		Control group		p-Value	Effect size
	Mean	(± SD)	Mean	(± SD)		
Noticing	3.11	(± 0.59)	3.08	(± 1.12)	0.866	N.A.
Not-distracting	2.47	(± 0.72)	3.76	(± 0.42)	0.001	2.313
Not-worrying	2.38	(± 0.89)	3.16	(± 0.68)	0.001	0.956
Attention regulation	2.18	(± 0.63)	2.92	(± 1,16)	0.001	0.738
Emotional awareness	2.87	(± 0.59)	2.99	(± 1.27)	0.516	N.A.
Self-regulation	1.83	(± 1.04)	2.50	(± 1.07)	0.004	0.636
Body listening	2.09	(± 1.11)	2.95	(± 1.33)	0.001	0,718
Trusting	1.47	(± 1.07)	3.32	(± 1.15)	0.001	1.668
MAIA total score	2.30	(± 0.42)	3.09	(± 0.78)	0.001	1.168

Abbreviation: MAIA, Multidimensional Assessment of Interoceptive Awareness; BD2-D, Bipolar II Disorder during depressive phase; ±SD, ± Standard Deviation; N.A., Not Applicable.

Secondly, analyzing interoceptive sensibility an intriguing pattern emerges. While individuals with BD2-D exhibit a preserved capacity for essential interoceptive detection – as reflected in comparable scores to healthy controls on the Noticing and Emotional Awareness subscales – this apparent preservation cover a deeper dysfunction. Indeed, significant differences emerged across dimensions responsible for integration and regulation of internal bodily signals, including Not-Distracting, Not-Worrying, Attention Regulation, Self-Regulation, Body Listening, and Trusting.

These components reflect higher-order processes that allow the brain to contextualize and adaptively interpret interoceptive input, transforming raw signals into coherent experiential and regulatory outcomes. The observed reductions suggest a failure not at the level of afferent signal reception, but at the level of integration, where the organism synthesizes internal signals into a stable and adaptive embodied self-model. In this sense, the interoceptive profile of BD2-D may be understood as a fall in the dynamic regulation of internal states, impairing the organism's ability to maintain affective homeostasis and respond coherently to internal perturbations.

Additionally, correlation analyses indicated that reduced interoceptive sensibility was generally associated with increased ED. Specifically, lower scores on the MAIA subscales, particularly those related to self-regulation and trust in bodily signals, were significantly correlated with greater difficulties in emotion regulation. This pattern suggests that individuals with BD2-D who struggle to interpret and regulate interoceptive signals also experience greater challenges in managing emotional responses.

The strongest correlations emerged between the self-regulation component of MAIA and the DERS components of Goals, Strategies, and Impulse, indicating that difficulties in accurately perceiving and integrating bodily signals contribute to difficulties in goal-directed behavior, reduced access to effective regulation strategies, and heightened impulsivity during emotional distress. Furthermore, since these correlations remained significant even after controlling for depressive and anxiety symptoms, the reduction in interoceptive sensibility appears to be a distinct factor influencing emotion regulation rather than a byproduct of general affective distress.

This was further confirmed by the mediation analysis, which demonstrated that BD2-D contributes to ED both directly and indirectly through the blunting of interoceptive sensibility. BD2-D had a strong direct effect on ED ($\beta = 21.52$, $p < 0.001$), accounting for 49.2 % of the variance in DERS scores. Additionally, BD also significantly reduced interoceptive sensibility ($\beta = -0.78$, $p < 0.001$), which in turn exacerbated ED ($\beta = 16.19$, BootCI [8.54, 27.55]). This suggests that reductions in interoceptive sensibility mediate a substantial portion of BD2-D's impact on emotional instability.

Notably, MAIA scores had a strong independent negative effect on ED ($\beta = -20.62$, $p < 0.001$), indicating that higher interoceptive sensibility is directly associated with lower ED, even when BD2-D is accounted for. These findings support the view that interoception acts as an active regulatory mechanism, serving as an intrinsic determinant of emotional stability.

Interoception plays a critical role in emotional regulation by enabling individuals to recognize, understand, and effectively respond to emotions through the awareness of bodily signals (Zamariola et al., 2019). Proper interoceptive sensibility aids in detecting early bodily changes associated with emotions, facilitating their regulation before

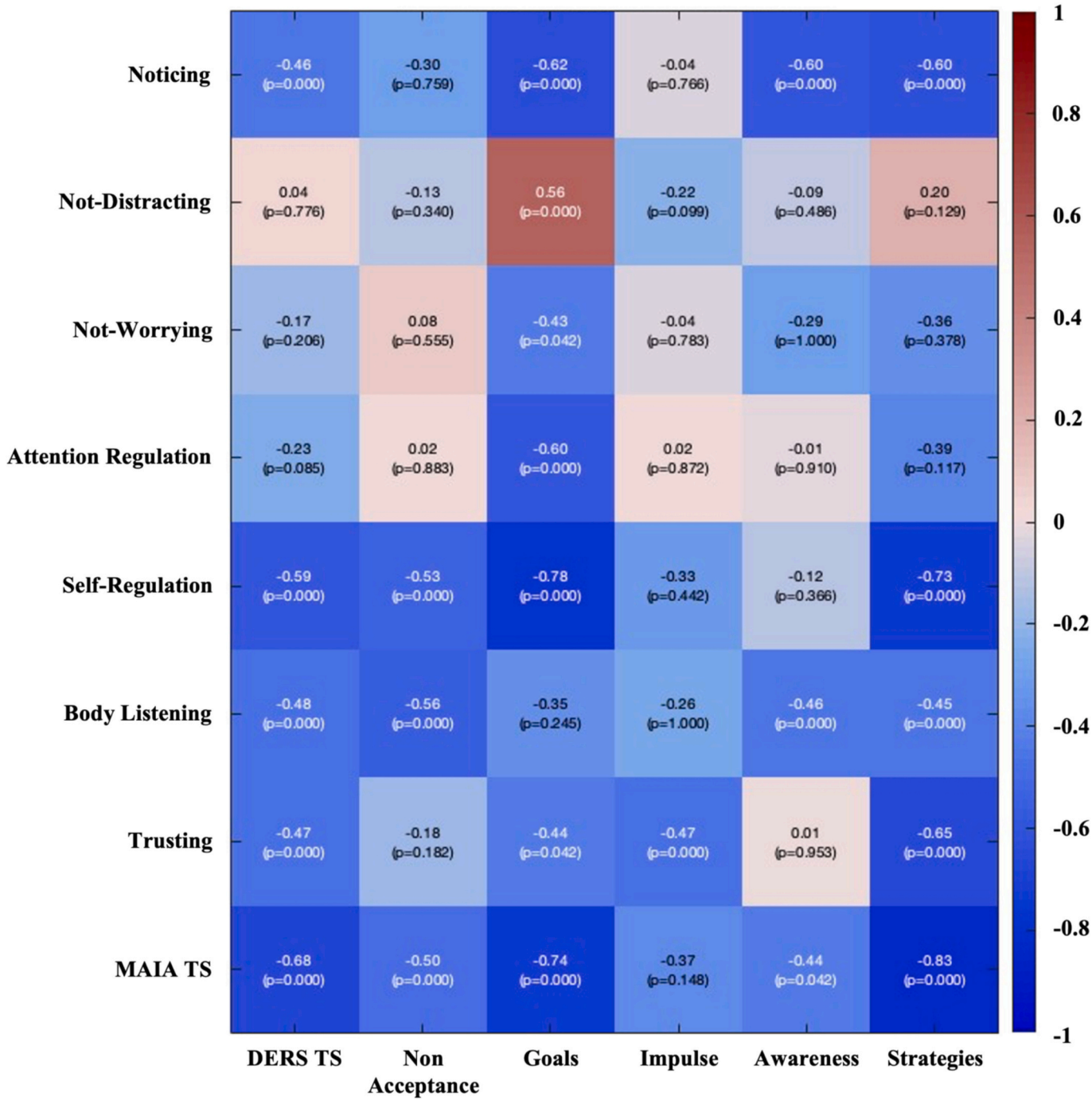

Fig. 2. Partial correlation matrix between interoceptive sensibility (MAIA) and emotion dysregulation (DERS), controlling for depression and anxiety symptoms. Note: Each cell displays the partial correlation coefficient along with the corresponding Bonferroni-adjusted p-value. The colour gradient represents the strength and direction of the correlations: blue indicates negative and red indicates positive correlations, with darker shades reflecting stronger associations (range: −1 to + 1). MAIA components are shown on the y-axis; DERS components are shown on the x-axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

they intensify; conversely, inaccurate interoceptive sensibility can make emotions feel confusing or difficult to distinguish, thereby increasing the risk of ED (Critchley and Garfinkel, 2017; Füstös et al., 2013; Zamariola et al., 2019).

Through the lens of the EPIC model, the brain continuously generates predictions about the body's physiological state and compares them with incoming interoceptive signals, thereby shaping subjective feeling states (Marshall et al., 2018; Seth, 2013). Self-representation emerges, therefore, from the minimization of prediction error between expected and actual interoceptive signals (Seth, 2013; Suzuki et al., 2013). However, when this finely tuned system falters, whether due to dysfunctional prediction updating or aberrant precision weighting of interoceptive signals, the organism struggles to effectively regulate internal states, leading to maladaptive emotional responses (Seth, 2013). From a neurobiological perspective, Perry and colleagues, introduced the concept of "interoceptive psychosis", proposing that in BD, widespread disruptions in neural networks responsible for processing and interpreting internal bodily signals - insula, anterior cingulate cortex,

Table 4
Mediation analysis.

Effect of IV→DV		R^2	β st	β	SE	p-Value	95 % CI	
							LLCI	ULCI
Total	BD2-D→ED	0.49		37.70	3.95	0.000	29.85	45.55
Direct	BD2-D→ED		0.82	21.51	3.63	0.000	14.30	28.72
Indirect	BD2-D→IS→ED		0.61	16.18	4.78	0.000	8.54	27.54
Effect of IV→MV	BD2-D→IS	0.30	−1.12	−0.78	0.12	0.000	−1.03	−0.53
Effect of MV→DV	IS→ED	0.21	−0.54	−20.62	2.53	0.000	−25.66	−15.58

Abbreviations: IV, Independent Variable; DV, Dependent Variable; MV, Mediating Variable; BD2-D, Bipolar II Disorder during depressive phase; ED, Emotion Dysregulation; IS, Interoceptive Sensibility; R^2, Coefficient of Determination; β st, Standardized Beta Coefficient; β, Beta Coefficient, SE, Standard Error; 95 % CI (LLCI-ULCI), 95 % Confidence Interval (Lower Level Confidence Interval-Lower Upper Confidence Interval).

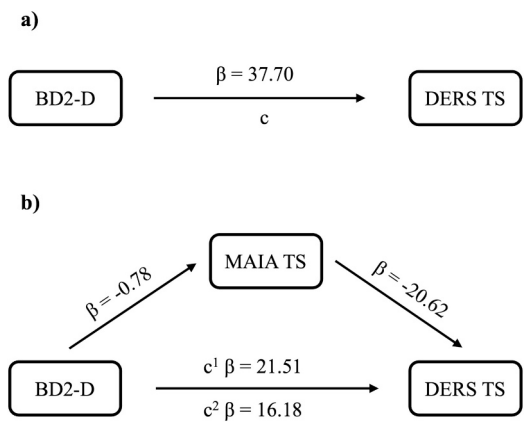

Fig. 3. Mediation model of the effect of BD2-D on Emotion Dysregulation via Interoceptive Sensibility.
Note:
(a) Total effect model: BD2-D significantly predicts increased emotional dysregulation (β = 37.70).
(b) Mediation model: BD2-D has both a direct effect on emotional dysregulation (c' β = 21.51) and an indirect effect (c^2 β = 16.18) via reduced interoceptive sensibility (β = −0.78). Lower interoceptive sensibility is associated with higher emotional dysregulation (β = −20.62). All paths are statistically significant ($p < 0.005$).
Abbreviations: BD2-D, Bipolar II Disorder during depressive phase; DERS TS, Emotion Dysregulation; MAIA TS, Interoceptive Sensibility; β, Unstandardized regression coefficient; c, Total effect; c^1, Direct effect; c^2, Indirect effect (mediation path).

orbitofrontal cortex, inferior frontal gyrus - result in interoceptive mismatch. This can lead inappropriate or exaggerated emotional responses, contributing to states of mania, depression and affective instability (Perry et al., 2019). As a matter of fact, emerging evidence suggests that the interoceptive network plays a crucial role in the pathogenesis of BD. Specifically, Qiu and colleagues identified altered connectivity patterns across distinct insular subregions during depressive episodes, indicating a disrupted integration of interoceptive, emotional, and cognitive processes (Qiu et al., 2020). Similarly, a meta-analysis revealed reduced resting-state insular activity in BD during depressive phase, suggesting compromised interoceptive and affective processing. This alteration may reflect dysfunctional integration of bodily and affective signals, potentially contributing to impairment in emotion regulation and increased mood instability (Pastrnak et al., 2021). It is worth noting that affective temperaments and chronic ED are known to significantly influence the course and expression of BD, acting as long-lasting modulators of affective instability (Akiskal and Akiskal, 2005). This suggests that the associations observed in this study, although focused on a specific phase, do not rule out the presence of other underlying trait vulnerabilities.

While further research is needed to refine these insights, our findings highlight the crucial role of interoceptive sensibility in shaping emotional regulation among individuals with BD2-D. These findings propose valuable opportunities for intervention: approaches that enhance interoceptive sensibility, such as mindfulness-based practices and body-focused therapies, may restore integrative capacity within the organism, supporting emotional stability and quality of life among individuals with BD2-D.

4.1. Study limitations

This study presents several limitations. First, although it explores a novel and promising area of investigation, the relatively small sample size together with the cross-sectional design limits the statistical power and generalizability of the findings. Second, although the aim of the study was not to investigate how BD2-D might alter interoceptive sensibility, it is important to consider the potential influence of pharmacological treatment in this regard. Previous research has shown that treatment with antipsychotics or mood stabilizers can affect the salience network (Martens et al., 2024) which may in turn impact interoceptive processing. We chose not to include pharmacological treatments as covariates in our analyses, as the relatively small sample size could have increased the risk of committing a Type II error. Third, all measures were taken during a depressive episode, reflecting state-level functioning rather than trait-level vulnerabilities. Future research should stratify patients based on baseline dysregulation traits and affective temperament profiles to determine whether the observed associations between interoceptive sensibility and ED persist across these stable characteristics. Fourth, although we reported the sample distribution by sex, we did not conduct stratified analyses by sex. Future studies should examine sex-related differences in the relationship between interoceptive sensibility and ED. Finally, the lack of more objective measures, such as functional magnetic resonance imaging or psychophysiological analysis, does not allow for a deeper understanding of the mechanisms involved.

5. Conclusion

In conclusion, our results highlight the pivotal role of interoceptive sensibility in mediating ED in BD2-D. This, underscore the importance of investigating interoception as a fundamental mechanism through which the organism shapes and regulates affective experience. Such insights could support the development of tailored interventions that enable individuals with BD2-D to better inhabit the shifting contours of their affective flow.

CRediT authorship contribution statement

Gaspare Alfi: Software, Methodology, Formal analysis, Data curation, Conceptualization, Writing – original draft. **Giovanna Grenno:** Methodology, Data curation, Writing – original draft. **Valerio Caruso:** Data curation, Writing – original draft. **Andrea Piarulli:** Software,

Formal analysis, Data curation. **Laura Palagini:** Supervision, Data curation, Conceptualization. **Giulio Perugi:** Supervision, Writing – review & editing. **Angelo Gemignani:** Supervision, Methodology, Conceptualization, Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare no conflict of interests.

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